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Soil formation and environmental reconstruction of a loess-paleosol sequence in Zmajevac, Croatia

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ABSTRACT

Loess-paleosol sequences are widely recognized in the Pannonian region from a sedimentological perspective; however, fewer studies are focusing on the soil formation processes within these paleosols. We present a detailed pedological study of one of the sequences in Zmajevac, aiming to complete the paleoenvironmental information related to these paleosols, particularly to decipher the nature and environmental conditions that formed the thick, slightly developed (*cumulus*) Bw horizons on top of the three paleosols in the selected sequence. Chemical, physical, mineralogical, and micromorphological analyses, as well as ^{14}C dating of shells, were performed on samples from the three paleosols and the recent soil of the Zmajevac sequence. The results suggest that a discontinuous but rather rapid supply of aeolian and partly alluvial materials led to weak soil formation in all paleosols, limited to calcium carbonate mobilization. The sedimentary accretion processes and milder climatic conditions account for the lack of significant rubefaction and the absence of clay illuviation. The most recent (ZN REC) soil is a paleoclimatological archive of the Bølling–Allerød interstadial period. Each older paleosol section represents three sedimentological and pedological events from MIS 3 (ZN 1), MIS 4 interglacial (ZN 2) and MIS 5e (ZN 3). In general, the younger the age of the horizons within each succession, the lower the level of development. These top horizons of investigated sequences are Bw (*cumulus*) horizons. Thus, this study enhances our understanding of paleoenvironmental conditions in the Pannonian region.

1. Introduction

Pedo-sedimentary complexes in eastern Croatia have been continuously studied for more than 150 years (Pilar, 1875; Gorjanović-Kramberger, 1912, 1922; Šandor, 1912; Mutić, 1990). In the last couple of decades, these sequences have been the focus of Quaternary investigations as archives of climate change in this Pannonian region (Bronger, 2003; Galović et al., 2009, 2011, 2023a, b; 2024; Molnár et al., 2010, 2011, 2021; Banak et al., 2012, 2013; Wacha et al 2013; Galović, 2014, 2016; Rubinić et al., 2015; Galović & Peh, 2016) (Fig. 1a). Based on these references, which use sedimentological, mineralogical, geochemical and statistical methods, the present state-of-the-art of their knowledge is the following:

(1) The regional climate of NE Croatia adjacent to the Danube River (from the Alps to the south of the Carpathian basin) resulted in loess accumulation during cold and arid periods of the last glacial period, with soil formation during interstadials and interglacials. In this area, the oldest infrared stimulated luminescence (IRSL) age estimates are 298 ± 24 ka (the end of OIS 10 and beginning of OIS 9, i.e. Middle Pleistocene) in the Šarengrad II section (Wacha et al., 2013; Molnar et al., 2021). Since the Vučedol Eneolithic

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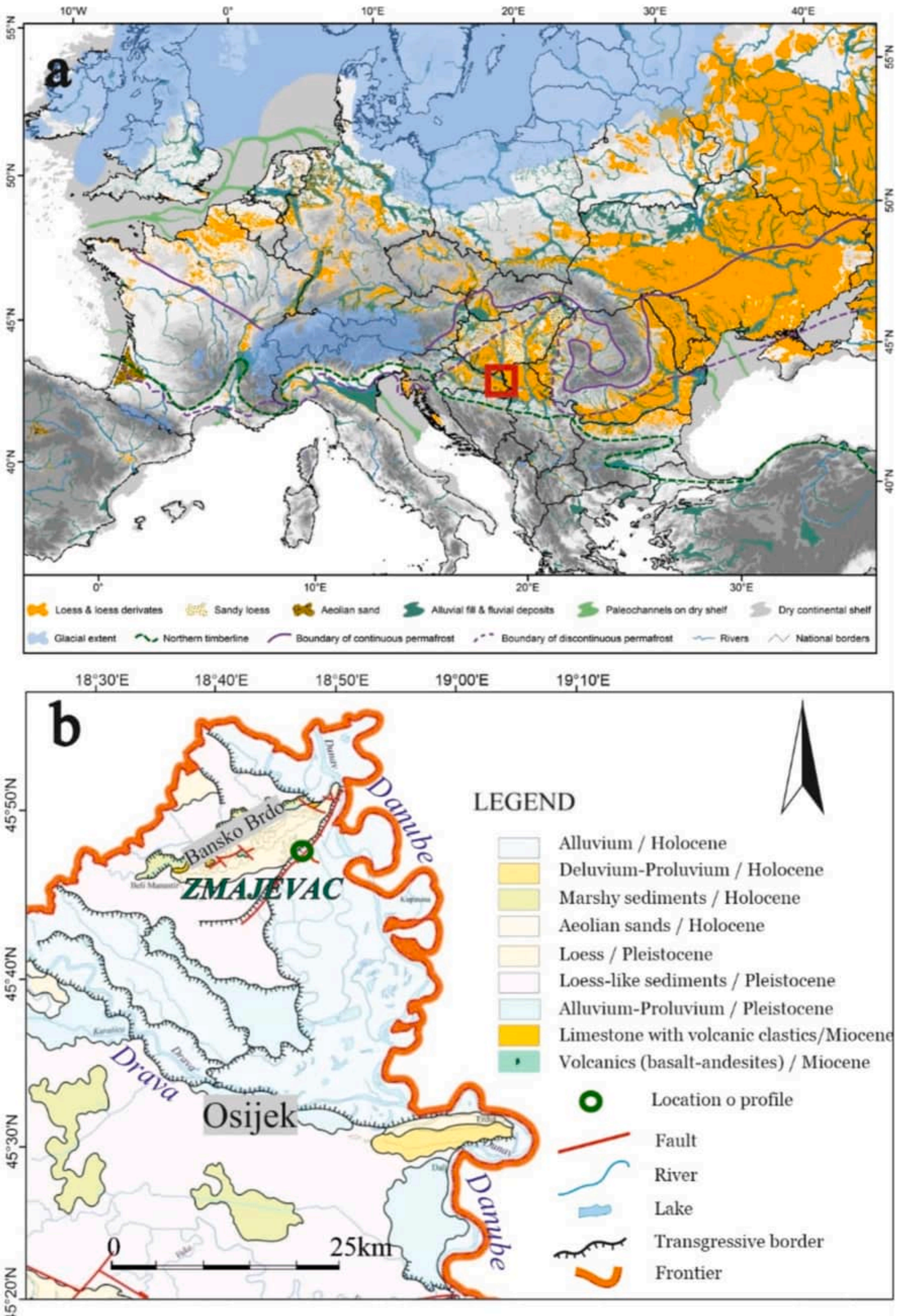

Fig. 1. (a) The position of the section in the European Loess Belt (modified from Lehmkuhl et al., 2021). Red rectangle marks the sampling location; (b) Detail of Geological map of Baranja. Green circle marks the sampling location (CGS-Department for Geology, 2009) showing the position of the section.

archaeological site (aged 3–2.2 ky BC) (Hincak et al., 2013) is covered with a couple of meters of loess (including one intercalated paleosol), the youngest loess in the area is from the historical period.

(2) The most important mechanisms of distribution of major and trace elements are clay illuviation and carbonate translocations, and are the main soil formation processes.
(3) The older paleosols are better developed indicating that during their pedogenesis the climate was more humid and warmer than in later warmer periods. Paleosol horizons could be separated based on weathering coefficients of Ba/Sr and ($CaO + Na_2O + MgO + K_2O)/Al_2O_3$.
(4) Warmer periods are not represented just by paleosols, but also by laminated alluvial sediments. During that period, large amounts of ice melted in the source areas, leading to a significant increase of available water mass and discharge.
(5) The source materials of these alluvial sediments were pedosediments developed in previously deposited (aeolian) material, which were reworked (transported and resedimented) by the high energy of oxygen-rich sediment flows (turbid water), regardless of whether and to what extent they were later exposed to post-sedimentary pedogenesis.
(6) Thick, very weakly developed paleosols (identified in the field as "cumulus" horizons) are overlying well-developed or weakly developed paleosols. They show a short-term pedogenesis was interrupted by their burial after deposition of the overlying loess. The significance of these paleosols as possible results of short warming periods at the beginning of glaciations has not yet been demonstrated, and represents a research gap that should be further clarified using a pedological approach.

Although these conclusions are assumptions based on geological investigations, recent surveys of modern soils in the Zmajevac area indicate that the current soils in the region are classified as Chernozems, according to both the Soil Classification of Croatia and the World Reference Base for Soil Resources (Galović et al., 2023a; IUSS-WRB, 2022). These authors identify loss of organic matter and moderate carbonate mobilisation as present-day processes taking place in these soils. They also point out that synpaleopedological aeolian sedimentation could have occurred, which would have slowed down soil formation by profile rejuvenation.

For a better insight into the processes and dynamics of climate change in the Pannonian region, it becomes necessary to perform a detailed pedological investigation of the paleopedological archive in one of the Zmajevac pedosedimentary complexes.

2. Objectives

The objectives of this research are:

a) To examine the impact of sedimentary and climatic processes on soil development in the Pannonian region.
b) To assess the extent of pedogenetic alteration of the parent material and its significance as a record of warm periods.
c) To decipher the soil formation processes, in particular calcium carbonate mobilization and possible clay illuviation.
d) To reconstruct the paleoenvironmental climatic conditions that affected paleosol formation, in particular to check whether the poorly developed, thick (Bw) horizons within the sequence are regular representatives of short warmer periods at the beginning of glaciations.
e) To analyse the stratigraphy of Quaternary deposits and their connections with global climatic changes.

To address these objectives, interdisciplinary approaches have been proved highly effective (e.g. Dodonov et al., 2006). In our study, we applied a range of methodologies to a loess-paleosol sequences in the Zmajevac region (Fig. 1b), which has been studied by combining different techniques with a pedological focus including their micromorphological, mineralogical and geochemical characteristics. Such multi-proxy approaches, which integrate diverse methods, are most effective in distinguishing between immediate and delayed responses to climate forcing.

3. Materials and methods

3.1. Geological setting

The Bansko Brdo (Bansko Hill) is the most prominent part of the relief in Baranja. This asymmetric elevation is a horst extending NE-SW and reaches 243 m above sea level. It is about 21 km long and up to 5 km wide. Tectonic movements uplifted the complex horst of The Bansko Brdo (Hećimović, 1991). This faulting revealed the profile of a steep slope along the south-eastern edge of The Bansko Brdo from Batina towards Kneževi vinogradi (Bognar, 1990).

The Bansko Brdo is the central morphogenetic unit of the SW-NE extension of Baranja (Fig. 1b). Its rise is indicated by the centrifugal type of the erosion network and the steep sides of the valleys where erosion prevails (Hećimović, 1991). Post-Miocene radial movements led to synsedimentary effusion of basalt-andesite in the form of plates and dikes and the deposition of volcanic breccias, as indicated by interstratified tuffs and montmorillonite clays, as a product of their alteration. The Badenian age of the bedrock andesite was confirmed by K-Ar dating (14.5 ± 0.4 and 13.8 ± 0.4 Ma at two locations) in Pamić & Pikija (1987) and Lugović et al. (1990). Pleistocene loess lies transgressively on them (Pikija et al., 1991; Bognar, 1990).

The borders with the surrounding river valleys are sometimes well-defined steep slopes (Pikija et al., 1991) formed by neotectonic activity until the youngest glacial period (Bognar, 1990). The neotectonic movements are still active today (Hećimović, 1991, Galović et al., 2009). On the southeastern edge of the Bansko Brdo, from Batina towards Kneževi vinogradi, there is probably a reverse basic longitudinal fault in the NE-SW direction (Grünthal & Stormeyer, 1983; Hećimović, 1991; Prelogović et al., 1998). The studied Zmajevac section is a loess-paleosol sequence exposed by tectonic uplift on the SE cliff of the Bansko Brdo Mt. (Fig. 1b). The sites, geological setting and stratigraphy are described in detail by Galović et al. (2009, 2011), Galović (2014, 2016), Molnár et al. (2010, 2011, 2021) and Wacha et al. (2021).

3.2. Climate

The climate of the region is temperate continental with dry summers, and the annual precipitation is about 700 mm (Jelić & Kalogjera, 2002; Croatian Meteorological and Hydrological Service,[2] Peel et al., 2007). The area differs from the rest of the central and western continental regimes in Croatia and is known as the driest part of Croatia (Perčec Tadić et al., 2023). NW and NE winds prevail at all sites. The mean annual wind speed is 4.6–4.8 ms^{-1} (Croatian Meteorological and Hydrological Service[1]).

3.3. Profile selection

The investigated section is situated at 45°48′49″N. Lat. and 18°49′33″ E. Long., at an elevation of 92 m asl. at the bottom of the section, about 2 km to the NE of the village of Zmajevac. The sediment succession has a thickness of 28.5 m. Three paleosols separated by not-weathered (unaltered) loess layers, fluvial sediments and the modern soil on top were distinguished and dated by sedimentological, geochemical and

[2] https://meteo.hr/klima.php?section=klima_hrvatska&param=k1_8; last accessed 21 August 2024.

mineralogical investigations by Galović et al. (2009) and Galović (2014) (Fig. 2). The whole section results therefore from a discrete relationship between paleosols and related sediments as proposed by Fedoroff et al. (2018).

3.4. Profile description

The lowest sediment layer in the sequence is silty loess, providing an IRSL age estimate of 217 ± 22 ka (OIS 8). The upper boundary is characterised by accumulations of gastropod detritus and carbonate-rich material forming lens-like structures. This horizon is then overlain by a brown pedocomplex (ZN3 paleosol), succeeded by more layers of loess. An IRSL age estimate of 121 ± 12 ka (OIS 6) was obtained from this layer, followed by another pedocomplex (double soil) intercalated with laminated sediment. Sharp discontinuities between horizons are marked by ripple marks and Fe/Mn-rich nodules. The entire succession is further covered by another layer of loess. Two samples extracted from this loess provided IRSL age estimates of 101 ± 10 and 68.6 ± 6.9 ka. Above the loess, a well-developed brown paleosol (ZN2 paleosol) from OIS 5 is found, followed by a less-developed paleosol (ZN1 paleosol) measuring about 205 cm in thickness, attributed to OIS 3. This paleosol layer is subsequently covered by loess and topped with modern rendzina soil (ZN REC modern soil). The youngest loess layer yielded IRSL age estimates ranging from 16.7 ± 1.8 to 20.2 ± 2.1 ka (OIS 3). For a more detailed lithostratigraphical subdivision of the profile, refer to Galović (2014). The three paleosols and recent soil previously identified were described according to pedological criteria (IUSS Working Group WRB, 2022).

3.5. Sampling methods

After removing half a metre of the outcrop to reduce the influence of weathering and vegetation, the horizons/layers were defined and described based on field observations (as colour, grain size, structure, texture, bioturbation, mottling, presence and form of carbonates; FAO, 2006). The horizons were primarily differentiated based on sedimentological characteristics, distinguishing between aeolian (homogeneous) and alluvial (laminated) horizons. The paleosols were then separated according to the extent of pedological development and within each paleosol, horizons were defined and grouped according to the degree of development (Retallack, 2001).

In each horizon, approximately 2 kg of bulk sediment was sampled from the upper to the lower boundary (Table 1, Fig. 2) to ensure that the sample represented an entire horizon (Galović et al., 2009; Galović, 2014).

Sixteen horizons, as shown in Fig. 2, were selected for chemical, physico-chemical and mineralogical analyses. The samples were taken from the whole horizons, whose depths from the surface are specified in Table 1 and Fig. 2. They were air-dried for about one month. The dried samples were sieved to the < 2 mm fraction to separate the sediment from larger carbonate concretions, while smaller concretions occasionally remained in the samples (Galović et al., 2011).

Besides, undisturbed samples in cylinders were taken from the same horizons for bulk density determination. Regarding micromorphological analyses, twelve undisturbed blocks, approximately 20x20x20 cm were carved and carefully packed for making soil thin sections (Fig. 3).

3.6. Physico-chemical methods

The pedological analyses were carried out in the laboratory of the Department of Soil Science at the University of Zagreb, Faculty of Agriculture. The types of individual soil analyses were carried out according to standard methods. Soil samples were prepared according to HRN ISO 11464 (2009): disturbed soil samples were air-dried, crushed, and sieved through the 2-mm-sieve.

3.6.1. Physical properties

Determination of soil particle size distribution (mechanical composition), i.e. fine earth, was carried out using the pipette-method, through wet sieving and sedimentation after soil dispersion by Na-

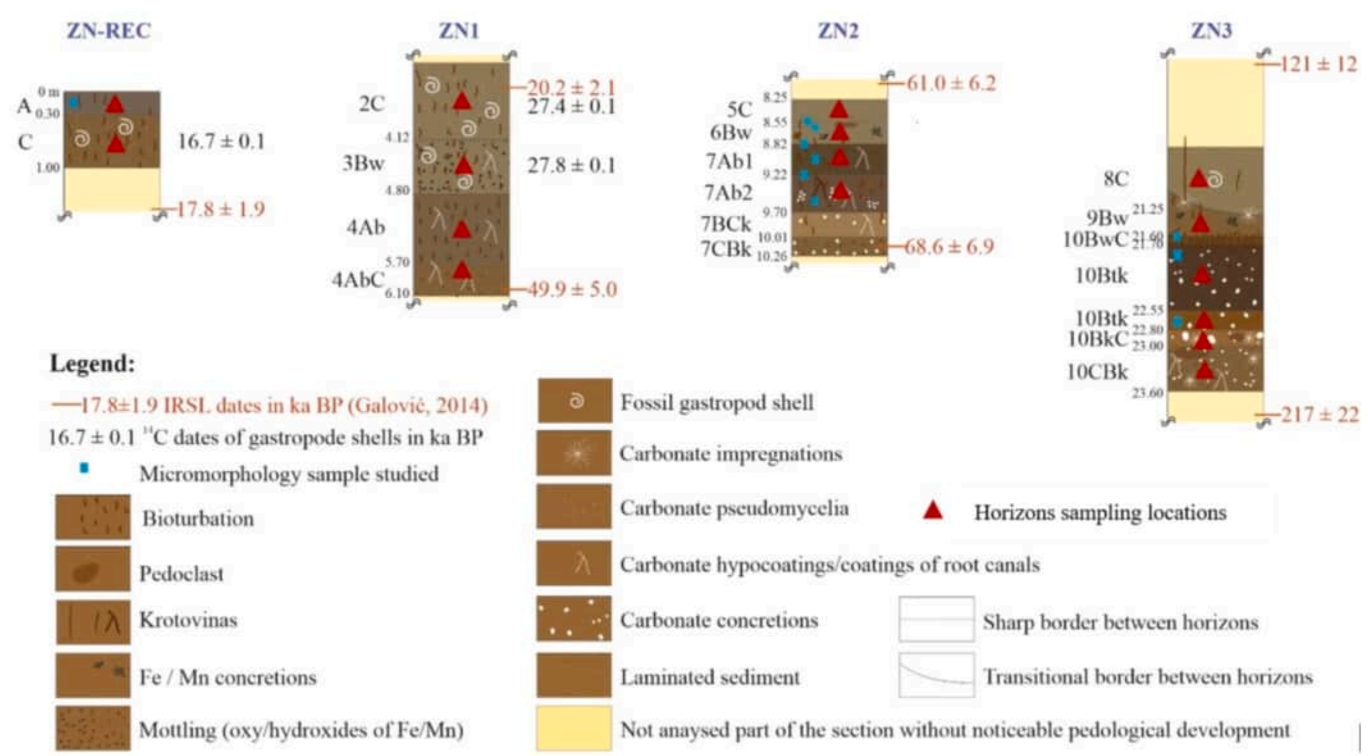

Fig. 2. Zmajevac pedosedimentary sequence, composed by a recent soil (ZN-REC) and three paleosols (ZN1, ZN2, ZN3) separated by unaltered loess. Depths (m) referred to the surface are indicated between the denomination of the genetic horizons. Dates in red correspond to IRSL datings from Galovič (2014).

Table 1
Summary of the profile description.

Profile	Horizon	Depth (cm)	Colour Munsell (moist)	Structure	Reaction to: * HCl	Reaction to: * H_2O_2	Pedofeatures
ZN-REC	A	0–30	10YR 5/2 (moist), 2.5Y 5/3 (dry)	crumb to granular	2	1	Bioturbation
	C	30–100	10YR 5/4 (moist), 2.5Y 6/2 (dry)	crumb to granular	2	0.5	Carbonate pseudomycelia, bioturbation
ZN1	2C	?-412	2.5Y 5/3	granular	2	0.5	Carbonate pseudomycelia, bioturbation
	3Bw	412–480	2.5Y 5/3	granular to blocky	2	0.5	Mottling (oxy/hydroxides of Fe/Mn), bioturbation, carbonate hypocoatings around root channels.
	4Ab	480–570	10YR 5/3	granular to blocky	2	0.5	Bioturbation, carbonate hypocoatings around root channels.
	4AbC	570–610	10YR 5/4	granular to blocky	2	0.5	Mottling (oxy/hydroxides of Fe/Mn), carbonate hypocoatings around root channels.
ZN2	5C	825–855	2.5Y 5/3	blocky to platy	2	0	
	6Bw	855–882	2.5Y 5/3	blocky	2	0–0.5	Mottling (oxy/hydroxides of Fe/Mn), bioturbation (inc. krotovinas), presence of a laminated, rounded clast (dim:20x7 cm) in the upper part of the horizon.
	7Ab1	882–922	10YR 4/3	blocky	2	0.5	Mottling (oxy/hydroxides of Fe/Mn), bioturbation, carbonate hypocoatings around root channels.
	7Ab2	922–970	7.5YR 4/3	granular	2	0.5	Bioturbation, carbonate hypocoatings and infillings in root channels, sub-rounded pedoclast dim. 5x10 cm, (upper border is sharp)
	7BCk	970–1001	10YR 6/4	granular to blocky	2	0	Carbonate concretions, infillings and hypocoatings, bioturbation.
	7CBk	1001–1026	10YR 5/4	crumb-granular	2	0	Carbonate concretions, mottling (oxy/hydroxides of Fe/Mn)
ZN3	8C	<2045–2125	2.5Y 5/3	laminar	2	1	Few krotovinas, gasteropod shell.
	9Bw	2125–2160	10YR 5/4	angular blocky to crumb	2	0.5–1	Krotovinas, bioturbation, concretions of Mn oxides
	10BwC	2160–2170	10YR 4/4	crumb to granular	2	0.5	Bioturbation
	10Btk	2170–2255	7.5YR 3/3	angular blocky	2	0.5–1	Mottling (oxy/hydroxides of Fe/Mn), bioturbation, carbonate concretions
	10Btk	2255–2280	7.5YR 4/6	blocky	2	0.5–1	Carbonate concretions
	10BkC	2280–2300	7.5YR 5/6	granular to blocky	2	0.5	Abundant carbonate concretions and impregnative soft nodules
	10CBk	2300–2360	10YR 5/4	granular	2	1	Mottling (small, oxy/hydroxides of Fe/Mn), bioturbation, carbonate concretions and impregnative hypocoatings. Fragment 6x17 cm, subhorizontal, color 5YR 4/4 (moist), corresponding to a pedoclast from an upper horizon

* 2: bursting bubbles, evident dissolution of the material; 1.5: big bubbles; 1: hearing, small bubbles; 0.5: only hearing.

pyrophosphate according to HRN ISO 11277 (2011) with evaluation of texture classes according to FAO (2006). Particle size distribution allowed us to calculate the Grain Size Index (GSI). According to Rousseau et al. (2007), this index is calculated as the ratio of the particles (20–50)μm/<20 μm. Since the particle sizes of FAO (2006) use 63 μm instead of 50 μm for the boundary between silt and sand, we have used the ratio coarse silt to fine silt and sand as an approximate value of GSI. The stability of structural microaggregates was determined according to the Vageler method (JDPZ, 1971; Škorić, 1982) i.e. by comparing the clay content determined in water and in Na-pyrophosphate. The determination of the volume density of dry soil was carried out gravimetrically according to HRN ISO 11272 (2004) using core samples of 100 cm^3. The determination of the density of solid particles was carried out in water, using the 100 mL pycnometer, according to HRN ISO 11508 (2004). The determination of water retention capacity of soil, total pore content in soil and air capacity of soil was performed according to the Gračanin method (JDPZ, 1971; Škorić, 1982) as follows: water retention capacity gravimetrically, by capillary water uptake of 100 cm^3 soil cores; total pore content by calculation based on the ratio between soil bulk density and particle density; air capacity by calculation based on total pore content and water retention capacity.

3.6.2. Chemical properties

Soil pH determination (H_2O and 1 M KCl) was performed according to HRN ISO 10390 (2005) in 1:5 soil: water suspensions. The determination of the carbonate content was performed according to HRN ISO 10693 (2004) i.e. gas-volumetrically, using the Scheibler calcimeter. Humus content was determined according to the Tjurin method (JDPZ, 1966) i.e. by acid–dichromate digestion followed by back-titration with Mohr-salt. The organic carbon content was calculated from the humus content using the Van Bemmelen conversion factor (humus content/ 1.724). The colour of the soil in the wet and dry state was determined according to the Munsell and Book, (2013) and the structure according to FAO (2006).

3.7. Mineralogy

The mineralogical composition of the soil samples of the fractions < 2 mm (bulk samples) and < 2 μm (clay fraction samples) was measured by X-ray powder diffraction (XRD) with a PANalytical X' Pert PRO diffractometer, equipped with a Cu-tube, graphite monochromator, and Pixel detector. More extended information about the methodology can be found in Supplementary Materials.

3.8. Micromorphology

They were dried and impregated with polyester resin before cutting and polishing, using oil as cooling agent. Vertical thin sections 90x120 cm and 30 μm thick were obtained, following the methodology of Benyarku & Stoops (2005). They were analysed and described using an Olympus BX51 petrographic microscope combining different magnifications and light sources (plane-polarized light, PPL; and cross-polarized light, XPL), following the guidelines of Stoops (2021). Fig. 3 displays the pictures of the micromorphological sampling. The sampling points are also shown in Fig. 2.

3.9. Geochemistry

The chemical composition of the samples was determined using the commercial ACME Analytical Laboratory. Major oxides were

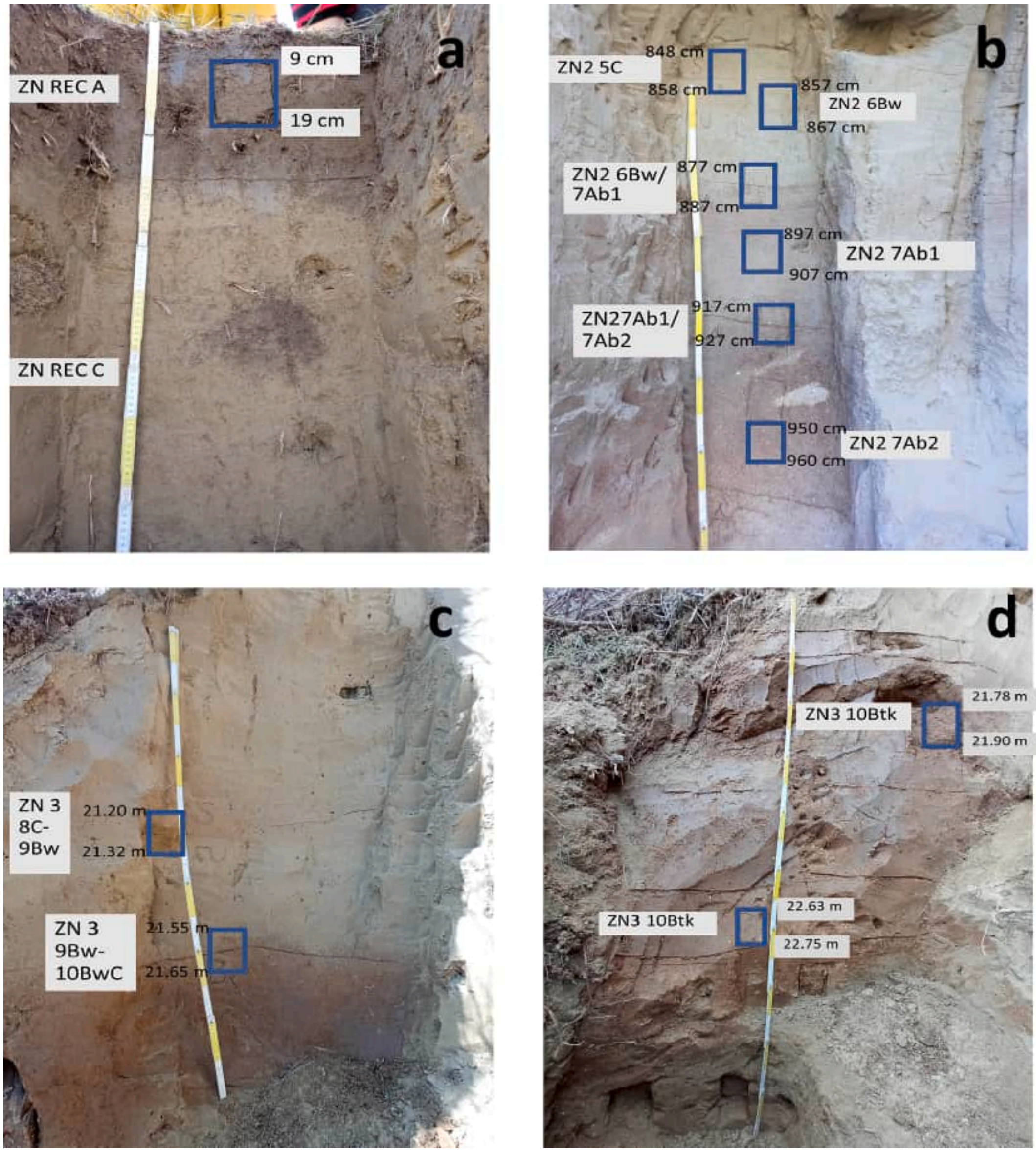

Fig. 3. Sampling points for micromorphology. (a) ZN REC; (b) ZN2, (c) ZN3-upper part; (d) ZN3-lower part.

determined using an X-ray fluorescence spectrometer (XRF) following $LiBO_2$ fusion. Extended information about the methodology can be found in Supplementary Materials. To approximate pedogenetic processes, weathering coefficients were determined to estimate the base loss [($CaO + Na_2O + MgO + K_2O)/Al_2O_3$], the leaching (Ba/Sr), the degree of feldspar weathering [($Na_2O + K_2O)/Al_2O_3$] and the ratio of plagioclase and K-feldspar weathering (Na_2O/K_2O) (Durn et al., 1999; Retallack, 2001; Buggle et al., 2008; Kahmann & Driese, 2008; Bokhorst et al., 2009; Galović, 2014; Galović et al., 2024).

3.10. Dating

In situ gastropod shells were extracted from the whole depth of horizons C (ZN REC), 2C and 3Bw (ZN1). They were pre-treated according to a standard acid-base-acid protocol, combusted to oxidize organic matter to CO_2, reduced back to graphite, pressed into the cathode and measured by accelerated mass spectrometry (AMS) in Direct AMS (Division of Accium Biosciences, Seattle, Washington, USA). A beam of C ions is produced by bombarding the surface of a graphite sample with Cs^+ ions. As such, the C beam is accelerated, focused and split into 14, 13 and 12 amu beams. The results of the AMS analyses have been corrected for isotopic fractionation with an unreported $\delta^{13}C$ value (Taylor, 1987) and presented in units of per cent modern carbon (pMC). Radiocarbon ages (0 BP = CE 1950) were calibrated with CALIB REV8.2 online software (Stuiver & Reimer, 1993), using Bayesian analysis and according to the IntCal20 calibration curve for the northern hemisphere (Reimer et al., 2020).

4. Results

4.1. Profile morphology and general physico-chemical characteristics

The main morphological characteristics of the profiles are displayed in Table 1. The colours (moist) of most of the horizons have hues between 10YR and 7.5YR, but the thick C and weakly developed Bw horizons on top of each sequence (“cumulus”) are yellower (2.5Y). They are calcareous throughout, with no signs of decarbonation; and the reaction to H_2O_2 is variable, normally agreeing with darker colours (accumulation of organic matter) or Mn-oxide (mottling due to paleoredox processes). Biological activity, in the form or galleries, root channels, infilled faunal channels (small krotovinas) and excrements are frequent. Besides, calcitic features are also very common, from fine pseudomycelia in the present soil to clear carbonate concretions in the bottom ZN3 sequence. Regarding structure, it is very variable, from granular (fine single grain) to crumb or blocky.

Table 2 shows the results of the physico-chemical analyses. All profiles have basic to alkaline reactions and are calcareous throughout, with calcium carbonate contents varying from 7 % to around 20 %. They tend to be higher in depth. Organic matter contents vary between 0.5 to 1 %, being highest in the buried A horizons. The textures are loamy to silty, with silt contents from 58 % to more than 83 %. Clay contents are maximum in the Bt horizons of ZN3 and the buried A horizons of ZN2, with increases of around 20 % with respect to the overlying horizons. In order to look for lithological discontinuities, the clay-free particle size distribution, together with the total granulometry and the Grain Size Index (GSI) in depth are displayed in Fig. 4. Most notably in the top horizons of each sequence, a distinct peak of the coarser fractions (coarse silt and fine sand) can be observed at 6Bw, 9Bw and 10Bkc, coinciding with the peaks of the GSI. In absence of the U-ratio values, which cannot be calculated from our data, these peaks would already indicate a higher energy deposition environment. Besides, the tren of coarse silt depth is clearly mirroring that of fine silt and clay, except in ZN3, which suggests that clay content in the overlying sequences is ruled by sedimentological rather than pedogenetic processes. Bulk density values reflect the degree of compactness of the horizons, from standard values for agriculture in the recent soil, to very compact in the bottom (ZN3) sequence.

4.2. *Mineralogy*

The bulk mineralogy, displayed in Table 3, is dominated by quartz, followed by plagioclase and mica in similar amounts, except in profile ZN3, where muscovite can reach up to 33 %. Carbonates are found in varying amounts in the form of calcite and dolomite. In addition, potassium feldspars, phyllosilicates, amphiboles, and, in some horizons, pyroxenes are also present. This mineralogical assemblage was found in previous investigations in this area (Galović, 2016; Grizelj et al., 2017; Urumović et al., 2017, Wacha et al., 2021).

Phyllosilicates make up a very small proportion of the total mineralogical composition of samples but they are very important features in explaining pedogenetic processes and reconstructing the environmental or paleoclimate conditions (Sheldon & Tabor, 2009). Semi-quantitative content of the fraction < 2 μm, also displayed in Table 3, determined that the most abundant clay minerals in all profiles are illite and chlorite. Their basal reflections at about 14 Å for chlorite and at 10 Å for illite did not change after each treatment. The peak at 7 Å is very pronounced on all diffractograms and it corresponds to chlorite and kaolinite.

Well-crystallized kaolinite is present in all samples except in ZN3 10BkC and ZN3 10CBk samples from the oldest paleosol profile. To determine whether kaolinite is present in paleosols, the oriented samples were saturated with DMSO. After this treatment, the peak of well-crystallized kaolinite from 7 Å shifted to 11.2 Å. Usually, in clay mineral identification, the peak of poorly crystalized kaolinite which is of pedogenetic origin remains near the 7 Å position after DMSO solvation (there is no shift to lower angles) (Velde & Meunier, 2008). In this case, it was not possible to determine whether poorly crystallized kaolinite was present in the paleosol profiles or not because the peak near the 7 Å position overlapped with the peak of chlorite. However, the results of modal analysis and petrographic descriptions of grains revealed the presence of feldspar grains with weathered rims that could consist of poorly crystallized kaolinite (Galović, 2016; Rubinić et al., 2015, 2018; Faivre et al., 2019; Galović et al., 2023a; Beerten et al., in press). Usually, it is very unlikely that poorly crystallized kaolinite can be found in loess and loess-derived soils. Even in some intensively weathered and rubified paleosols of the Middle and Early Pleistocene age, no, or only very small amounts of pedogenetic kaolinite were found (Brogner & Heinkele, 1993).

Smectite was recognized only in the few horizons by the expansion of the broad peak between 11 and 14 Å after ethylene glycol solvation of Mg and K-saturated samples and after glycerol solvation of Mg-saturated samples. Mostly it is part of the mineral assemblage of C and Bw horizons in recent and ZN1 paleosols. It is very likely interstratified with illite which makes a broad peak from 10 to 13 Å but also in some horizons, it is interstratified with chlorite. This is indicated by a high-angle shoulder near the peak of 14 Å after heating at 550 °C.

Vermiculite was recognized in diffractograms of K-saturated samples where its 14 Å peak shifted close to 10 Å. In some samples, the 002 peak of vermiculite at about 4.82 Å was recognized. It is the main component of the ZN2 and ZN3 profiles. In all paleosols, near the position of vermiculite, it is also recognized as hydroxy interlayered vermiculite (HIV), by the progressive collapse of 14 Å peak to ~10 Å after heating at 400 °C and 550 °C (Velde & Meunier, 2008; Huang et al, 2018). Huang et al (2018) suggest the formation of HIV by the polymerization of Al/Fe-hydroxy materials within the interlayer space of vermiculite and its further transformation to chlorite (pedogenetic chlorite with a peak at

Table 2
Main physico-chemical characteristics of the paleosols.

Profile	Horizon	pH (1:2.5)		$CaCO_3$ %	Organic Matter %	Particle size distribution (weight %)					Texture	Bulk density (Mg m^{-3})	Particle density (Mg m^{-3})	Porosity (%)
		H_2O	KCl 1 M			Coarse sand 2.0–0.2 mm	Fine sand 200–63 μm	Coarse silt 63–20 μm	Fine silt 20–2 μm	Clay < 2 μm				
ZN-REC	A	7.91	7.56	14.5	2.12	0.5	6.4	51.7	28.4	13.0	SiL	1.25	2.65	53
	C	8.38	7.94	20.4	0.80	0.3	7.5	51.6	28.3	12.3	SiL	1.21	2.70	55
ZN1	2C	8.97	8.20	20.2	0.54	0.3	7.1	57.0	26.4	9.2	Si			
	3Bw	9.01	8.18	15.4	0.59	0.2	5.7	54.5	27.1	12.5	SiL	1.24	2.74	55
	4Ab	9.03	8.16	7.7	0.72	0.4	7.7	51.9	26.2	13.8	SiL	1.26	2.75	54
	4AbC	8.91	8.10	10.2	0.78	0.2	6.6	50.1	26.0	17.1	SiL	1.39	2.74	49
ZN2	5C	8.75	8.19	16.2	0.47	1.2	10.2	63.2	19.2	6.2	L			
	6Bw	8.67	8.11	14.5	0.49	1.3	6.8	60.0	21.7	10.2	L	1.25	2.71	48
	7Ab1	8.61	7.87	11.8	0.96	0.3	4.1	48.3	24.4	22.9	L	1.27	2.67	48
	7Ab2	8.59	7.90	19.3	0.98	0.2	3.4	40.8	24.2	31.4	CL	1.12	2.67	50
ZN3	8C	8.80	7.77	14.1	0.36	0.4	8.6	60.1	22.9	8.0	Si			
	9Bw	8.31	7.56	9.1	0.52	2.0	5.8	52.0	26.5	13.7	SiL	1.42	2.72	48
	10Btk upper	8.58	7.46	8.5	0.80	1.9	6.2	39.3	19.7	32.9	CL	1.38	2.65	48
	10Btk lower	8.48	7.50	12.3	0.85	0.5	7.9	39.4	18.6	33.6	CL	1.32	2.65	50
	10BkC	8.49	7.55	18.5	0.59	2.5	6.0	37.1	23.3	31.1	CL	1.48	2.65	44
	10CBk	8.53	7.50	20.8	0.54	2.8	7.9	38.8	24.0	26.5	SiL	1.51	2.67	43

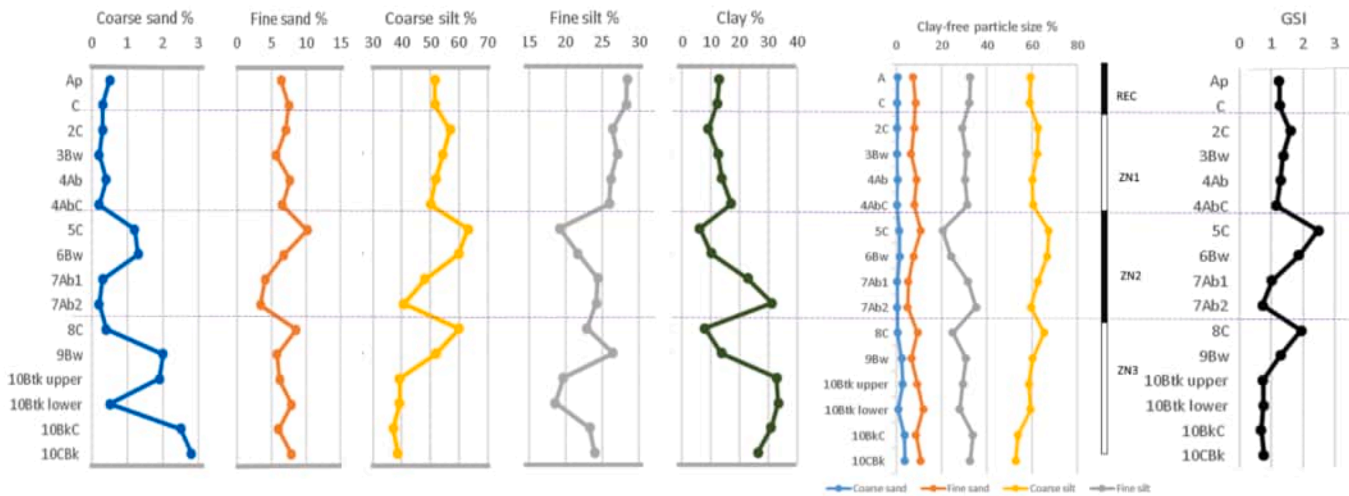

Fig. 4. Total particle size, clay-free particle size distribution and Grain Size Index (GSI) in depth.

~13.8 Å). It could be also the case in Zmajevac paleosols horizons because most of the identified chlorites in samples have a peak at ~13.8 Å. Except in this way, pedogenetic chlorite in Zmajevac was formed due to the transformation of micas (biotite, muscovite) which is proved by modal analysis (Galović, 2016; Galović et al., 2023a).

In addition, in all horizons, the presence of low-charge vermiculite (LCV) or high-charge smectite (HCS) was detected. Namely, after treatment of the Mg-saturated samples with ethylene glycol or glycerol, the 14 Å reflections were found to expand to 17–18 Å, but this effect was not visible in the reflections after ethylene glycol solvation of the K-saturated samples. It means that in all paleosol horizons a clay mineral that has a characteristic of both vermiculite and smectite is present (Moore & Raynolds, 1997; Durn et al., 1999; Terhorst et al., 2012). According to Egashira & Iwashita (1992) this kind of expandible clay minerals is the result of mica transformation.

Besides phyllosilicates, other minerals such as quartz, K-feldspars, plagioclase and goethite were found in the clay mineral fraction. The quartz content is constant along all paleosol profiles, while other mineral contents, such as goethite, vary along profiles. The presence of goethite is mostly evidenced in the horizons of ZN REC, ZN1 and ZN2 profiles. In other profiles, it is present in traces. It is very similar in appearance with potassium and feldspar content. These variations can be one of the indicators of the soil weathering degree.

4.3. Micromorphology

A summary of the micromorphological descriptions is found in Table 4. The microstructure of the studied horizons, as seen in thin sections, is generally apedal, except in some cases where it shows a weak subangular blocky structure. The main pores are either channels and vughs, or single packing pores between coarse silt grains where micromass is almost absent, as in the 6Bw horizon of ZN2. Accordingly, the c/f related distribution is either fine monic or close porphyric. The coarse fraction is made of very fine sand and coarse silt of quartz and feldspars, calcite and limestone grains, muscovite flakes, and occasionally fragments of shells. The 6Bw horizon of ZN2 shows a banded distribution of the particles with a clear sorting (Fig. 3a). The b-fabric of the micromass is mostly crystallytic micritic due to the presence of micrite, although in those horizons with less carbonates, it becomes stipple–speckled. The juxtaposition of both fabrics in some of the horizons (as in 6Bw-7Ab1 and 7Ab2 of ZN2 and 10BwC of ZN3) could be due to horizon mixing by bioturbation, as passage features by fauna shown by the crescent arrangement of the particles is evident throughout all the profiles. Organic components are almost absent, except for a few root residues.

Pedofeatures are either amorphous/cryptocrystalline or calcitic. Surprisingly, no textural pedofeatures, as clay coatings of any type are observed in the clay-rich (Bt) horizons, pointing to a low pedogenesis degree and to an original clay-rich sediment, since clear clay neoformation has not been detected as a distinct clay mineralogy trend. Amorphous pedofeatures are present almost in all horizons and consist of impregnative nodules of Fe- and Mn-oxihydroxides, aggregated, orthic, from faint to distinct, and in some cases concentric (profile ZN3, horizons 10Btk) (Fig. 5b, c, d). They are only absent in the 7Ab2 and 6Bw horizons of profile ZN2.

Calcitic features are very frequent, and abundant in ZN3. They consist of micrite nodules, rounded and irregular, orthic, and disorthic (Fig. 5e); loose discontinuous infillings and coatings of needle calcite (Fig. 6a); impregnative micritic hypocoatings around channels, infilled by decarbonated micromass, some of them with root remains; and impregnative hypocoatings of micrite around channels (Fig. 6b). Biogenic calcite is also frequent, as queras (Herrero et al., 1992), which are infillings of calcite pseudomorphs after apical cells, although partly disturbed and micritised (Fig. 6c, d); also rhyzoliths (calcified roots), and earthworm spherulites (Fig. 6e).

4.4. Geochemistry

The distribution of SiO_2, Fe_2O_3 and Al_2O_3 is similar through all profiles. The minimum enrichment in these elements is found in sample ZN2-7BCk, where CaO shows the highest value. MgO and CaO show similar patterns, also observed for K_2O and Na_2O (Supplementary Table 1). These values allowed us to calculate different geochemical ratios, which are displayed in Fig. 7.

$CaO + Na_2O + MgO + K_2O)/Al_2O_3$ is a coefficient that yields qualitative information about the loss of bases. The coefficient varies significantly, with values ranging from 1.7 to 7.9. The maximum value is observed in the ZN2-7BCk horizon.

The Ba/Sr ratio is a leaching coefficient that represents resistance to drainage (estimation of leaching). It is commonly used to reflect the chemical weathering intensity, due to the solution rate diversity between Sr and Ba during pedogenic processes (Liang et al., 2013). Ba is a resistant element with lower mobility, and Sr is very mobile in a pedogenetic environment. This coefficient is highest in the ZN3-10Btk horizon, and lowest in the ZN2-AC horizon.

Opposite trends between $CaO + Na_2O + MgO + K_2O)/Al_2O_3$ (increasing in depth) and Ba/Sr (decreasing in depth) ratios are observed in ZN1, and the upper part of ZN2; and the reverse (decreasing and increasing respectively) in ZN3; which would indicate gain / losses of bases as the main processes occurring in the sequence.

Finally, the Na_2O/K_2O and the $(Na_2O + K_2O)/Al_2O_3$ ratios are

Table 3
Mineralogical composition of the studied profiles.

PROFILE	Horizon	Mineralogical composition of fraction < 2 mm									Mineralogical composition of fraction < 2 µm											
		Qtz	Cal	Dol	Pl	K-Fs	Am	Px	Phyl	Ms	Kln_D	Chl	Sm	Vr	HIV	HCS/LCV	MLCM	Il	Qtz	Gt	K-Fs	Pl
ZN REC	**A**	30	12	15	12	11	2	–	6	12	x	xx	–	–	–	x	x	xxx	x	x	x	x
	C	29	18	11	14	6	2	–	6	13	xx	xxx	x	~	–	x	x	xxx	x	x	x	x
ZN 1	**2C**	30	7	16	17	8	1	–	9	12	x	xx	x	–	x	x	x	xxx	x	x	x	x
	3Bw	30	3	15	17	12	1	–	6	16	x	xx	x	~	x	x	x	xxx	x	x	x	x
	4Ab	30	3	8	20	9	2	~	9	19	x	xx	–	–	–	x	x	xxx	x	x	x	x
	4AbC	29	8	6	16	12	1	–	9	20	x	xx	~	–	–	x	x	xxx	~	~	x	x
ZN 2	**5C**	25	11	17	16	8	1	–	7	15	x	xx	~	x	x	x	x	xxx	x		x	x
	6Bw	26	10	14	14	14	2	–	7	13	x	xxx	~	xx	x	x	x	xxx	x	~	x	x
	7Ab1	33	8	7	15	8	1	–	9	19	x	xx	–	x	x	x	x	xxx	x	x	~	~
	7Ab2	34	16	3	14	2	1	–	6	24	x	xx	–	–	x	x	x	xxx	x	x	~	x
	7BCk	27	31	11	11	1	~	–	3	16	x	xx	–	–	–	x	x	xxx	x	x	~	~
ZN 3	**8C**	28	7	11	21	2	1	1	8	21	xx	xx	–	x	–	x	x	xxx	x	x	~	x
	9Bw	30	4	7	16	13	1	–	11	17	xx	xxx	–	x	x	x	x	xxx	x	–	–	~
	10Btk upper	35	10	~	13	6	1	~	6	29	x	xx	–	–	?	x	x	xxx	~	–	~	~
	10Btk lower	33	10	~	15	13	1	–	7	21	x	xx	–	x	x	x	x	xxx	x	~	~	~
	10BkC	31	14	~	8	18	–	–	3	26	–	xx	–	–	?	x	x	xxx	x	~	~	~
	10CBk	29	18	2	13	1	–	–	5	33	–	xx	–	–	?	x	x	xxx	x	–	~	x

Legend: Am-amphibole, Cal-calcite, Chl-chlorite, Dol-dolomite, Gt-goethite, HCS/LCV-high charge smectite or low charge vermiculite, HIV-hidroxyinterlayerd vermiculite, Il-illite, Kln_D-well crystalized kaolinite (kaolinite solvated with DMSO) K-Fs-potassium feldspar, MLCM-mixed-layered clay minerals, Ms-muscovite, Phyl-phyllosilicates, Pl-plagioclase, Px-pyroxene, Qtz-quartz, Sm-smectite, Vr-vermiculite, ~ – in traces,? – presence with uncertainty, x – relative abundance of minerals within horizons based on X-ray diffraction (no quantitative value is assigned to x).

Table 4
Summary of the micromorphological descriptions.

Pro-file	Horizon	c/f ratio	c/f related distribution	Microstructure	Poro-sity	Pedofeatures					
						Nodules of Fe/Mn- oxi-hydroxides	Passage features	Calcitic pedofeatures			
								Micrite nodules	Needle calcite	Hypo-coatings	Biogenic calcite
ZN REC	A	1/1	close porphyric	channel / vughy	30 %	x	xx	x			
ZN1	2C-3Bw	1/1	close porphyric	channel / vughy	20 %	x	xx				
ZN2	5C-6Bw		monic	single grain	30 %	x	x			x	
	6Bw		monic	single grain	25 %					x	
	6Bw-7Ab1	3/1	monic-enaulic-close porphyric	channel	30 %	x				x	x
	7Ab1	1/1	close porphyric	weak subangular blocky, almost massive	20 %	x	x	x	x		x
	7Ab2	2/3	close porphyric	channel / vughy	30 %			x	x		x
	7Ab1-7Ab2	1/1	close porphyric	very weak coarse granular, intapedal channel / vughy	25 %	x	xx	x			x
ZN3	10BwC upper	4/1	close porphyric to monic	channel	35 %	x		x		x	x
	10BwC lower	1/1	close porphyric	channel / vughy	20 %	x	xx	x		xx	x
	10Btk upper	1/1	close porphyric	very weak subangular blocky, intapedal channel	15 %	x	xx	x		x	x
	10Btk lower	2/3	single space porphyric	channel / vughy	20 %	x	xx	xx	x	x	x

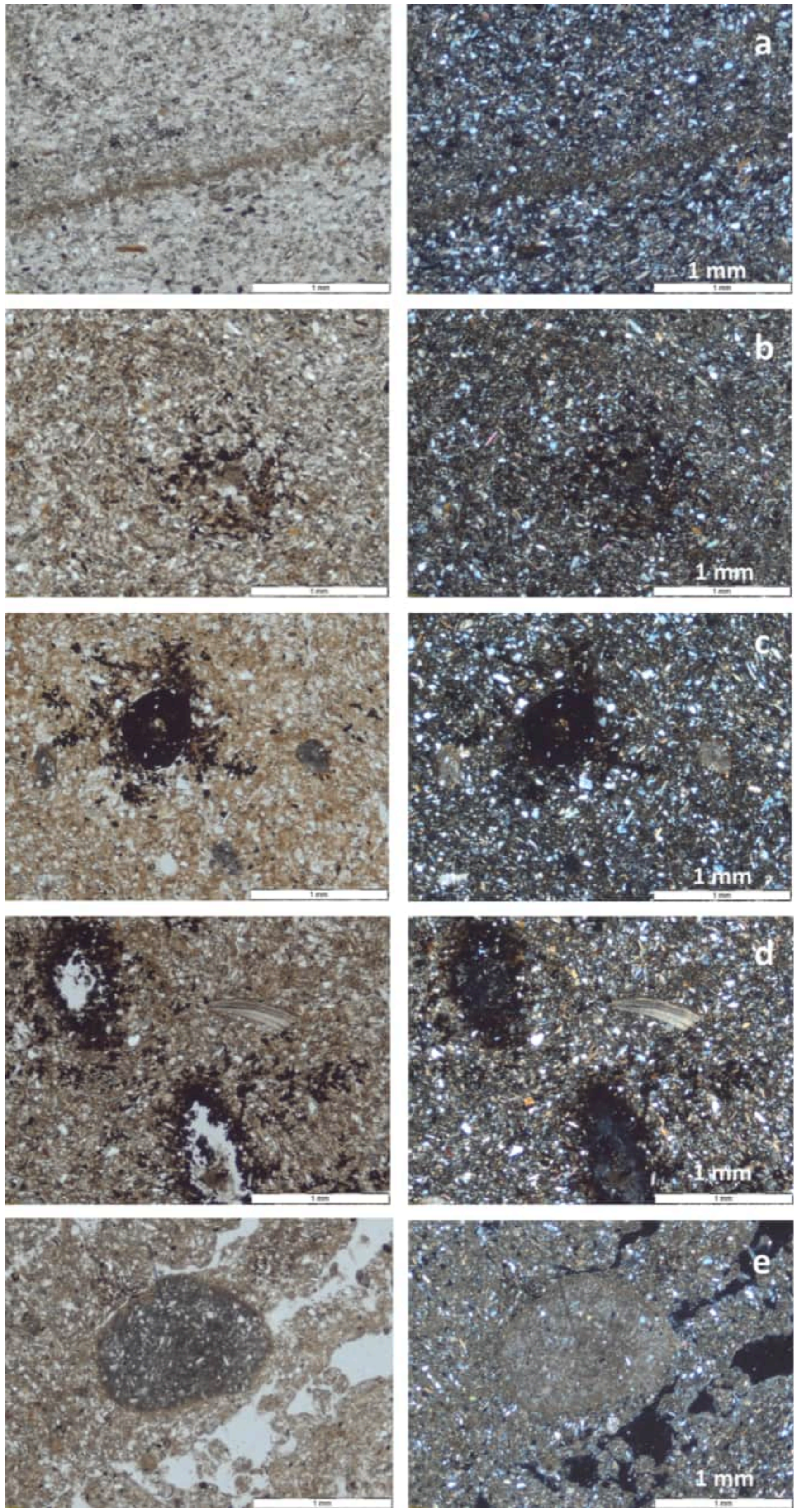

Fig. 5. Microphotographs of selected features (left: parallel polarizers, right: crossed polarizers). (a) Sorted layer (sedimentary feature), horizon 6Bw, profile ZN2. (b) Aggregated impregnative nodule of Mn oxides, horizon 6Bw-7Ab1, profile ZN2. (c) Concentric impregnative nodule of Mn oxides, horizon 10Btk, profile ZN3. (d) Impregnative hypocoatings of Mn-oxides, and shell fragment, horizon 7Ab2, profile ZN2. (e) Impregnative disorthic nodule of $CaCO_3$ and biopores infilled with excrements, horizon 7Ab2, profile ZN2.

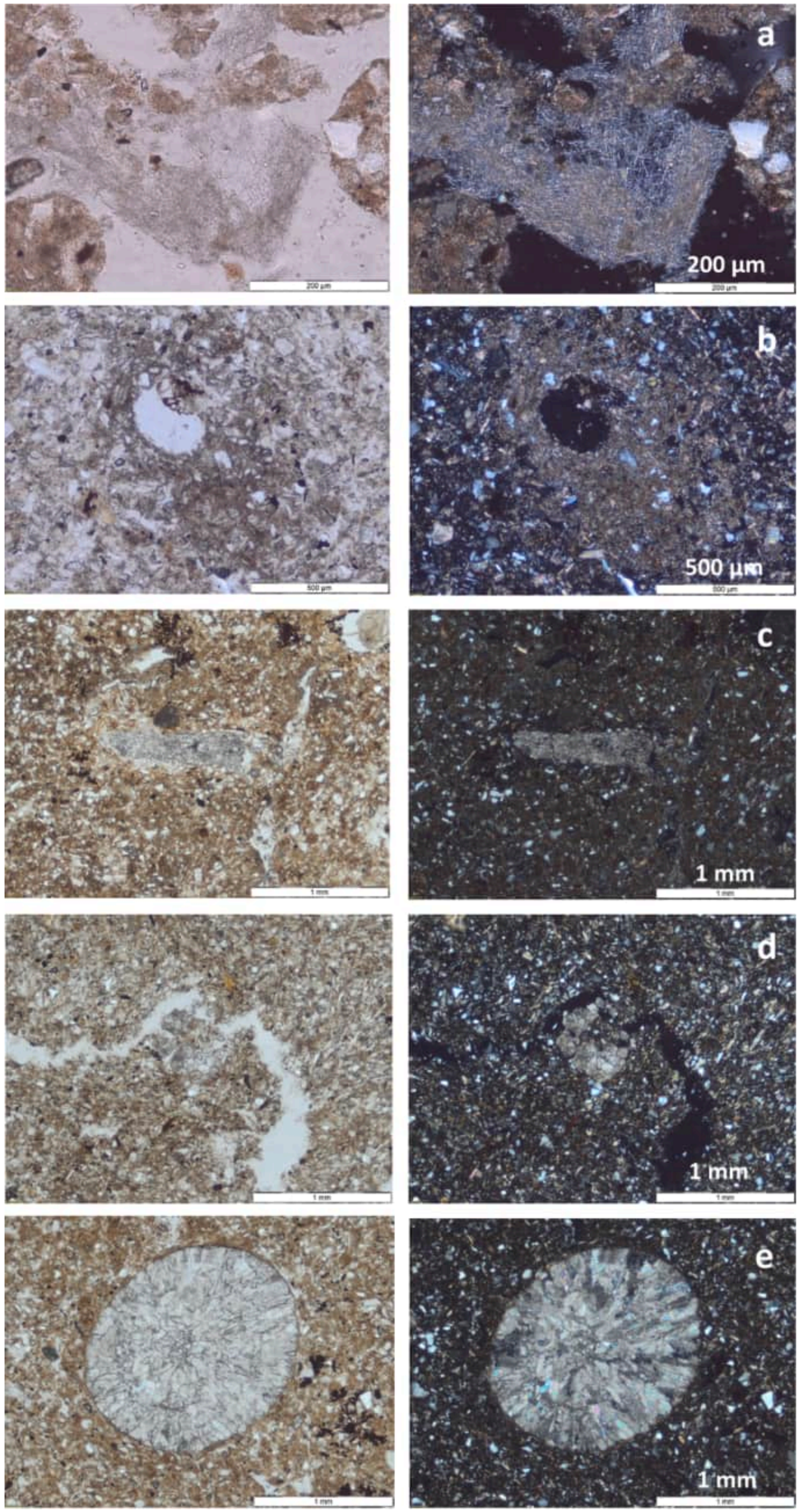

Fig. 6. Microphotographs of selected features (left: parallel polarizers, right: crossed polarizers). (a) Discontinuous loose infilling of needle calcite, horizon 7Ab2, profile ZN2. (b) Impregnative hypocoating of micrite around a biopore. Horizon 6Bw-7Ab1, profile ZN2. (c) Dense complete infilling of biosparite (quera). Horizon 10Btk, profile ZN3. (d) Nodule of biosparite crystals (fragment of quera). Horizon 7Ab2, profile ZN2. (e) Calcareous earthworm spherulite, horizon 10Btk, profile ZN3.

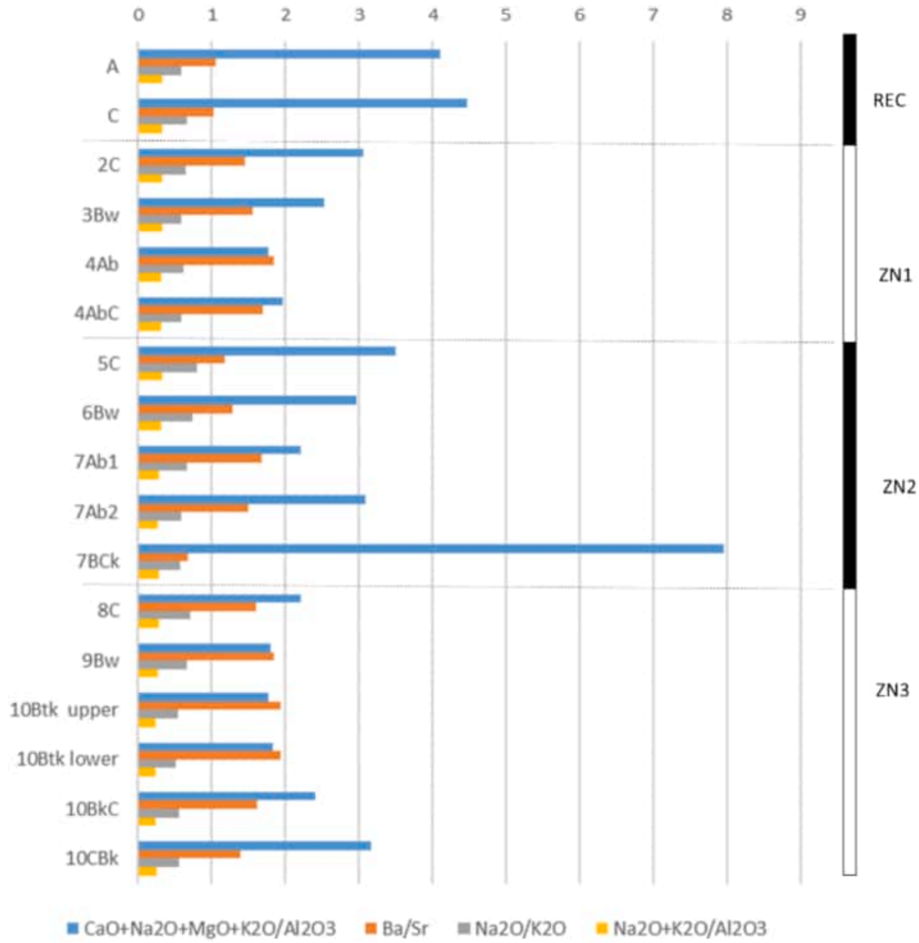

Fig. 7. Distribution of the geochemical ratios of the analysed horizons.

weathering coefficients related to the ratio of plagioclase and K-feldspar, and the degree of feldspar weathering respectively. Based on a homogenous mineral composition (Galović, 2016), a low Na_2O/K_2O ratio indicates an intense weathering of less resistant plagioclases and a higher degree of pedogenesis. The highest value of this coefficient is found in the ZN2-5C horizon, while the lowest one is in the ZN3-10Btk lower horizon. The $(Na_2O + K_2O)/Al_2O_3$ ratio represents the resistance of feldspars compared to resistant Al-oxides, indicating the intensity of pedogenesis, since feldspars are the dominant source material for clay minerals. In all investigated horizons, the value is generally greater than 0.1, with the highest values in the ZN1 profile. In spite of their variability, both coefficients show rather homogeneous values and no clear trends, indicating a low degree of both mineral weathering and clay formation.

The analysed concentrations of trace elements, like Co, W, Y, Nb, Sn, Cs, Ni, Rb, Zn, V, Ga, Hf, Th, Zr, Ba and As, have similar distributions along the investigated profiles with an exception for Cu and Pb. The specific distribution of Sr concentrations follows the distribution of Ca because of their isomorphic exchange in carbonates (Supplementary Table 2). Fig. 8 shows the trends of the most abundant trace elements, where the particular distribution of Sr is observed as mirroring that of the rest of the elements. The high amount of Cu at the surface of the recent soil should be attributed to fertilisation. Minimum concentrations are observed for sample ZN2-7BCk, and the maximum occurs in the ZN3 profile (Supplementary Table 3).

[Supplementary Table 1 and 2. Total element content (oxides and trace elements)].

[Supplementary Table 3. Elemental statistics of the geochemistry analyses.].

4.5. Dating

The radiocarbon dating results are presented in Table 5 in units of per cent modern carbon (pMC) and the uncalibrated radiocarbon age before the present (BP; 0 BP = CE 1950). All results have been corrected for isotopic fractionation with an unreported $\delta^{13}C$ value. The calibrated age of gastropod shells from sample ZN-REC C is 16.704 ± 116 cal BP. Sediment overlying ZN1 (ZN1 2C) is dated 27.366 ± 113 cal BP and the horizon below (ZN1 3Bw) is dated 27.849 ± 89 cal BP, respectively (Table 5, Fig. 2). The dating of gastropod shells matches with previously published IRLS dating results of the loess (Galović et al., 2009).

5. Discussion

5.1. Main soil formation processes

The four sequences within the profile identified by Galović (2014) in the Zmajevac profile are corroborated, on one hand, by the trend of the base loss $((CaO + Na_2O + MgO + K_2O)/Al_2O_3)$ index, which tends to decrease towards the top in sequences ZN1 and upper part of ZN2, in such a way that the least developed horizons (“cumulus” Bw and C) tend to have the lowest values. These trends are apparently reversed in the deepest ZN3 section, which could correspond to recarbonatation from the overlying material. In contrast, the Ba/Sr (leaching, resistance to drainage) ratios show the reverse trend, as observed by Galović (2014). Based on $(Na_2O + K_2O)/Al_2O_3$ ratio, feldspar weathering is the highest in the ZN1 profile. The obtained results completely match both the bulk and clay mineralogy. Durn et al. (2018) found the same trend of this ratio in red paleosols. The leaching of basic cations agrees with the

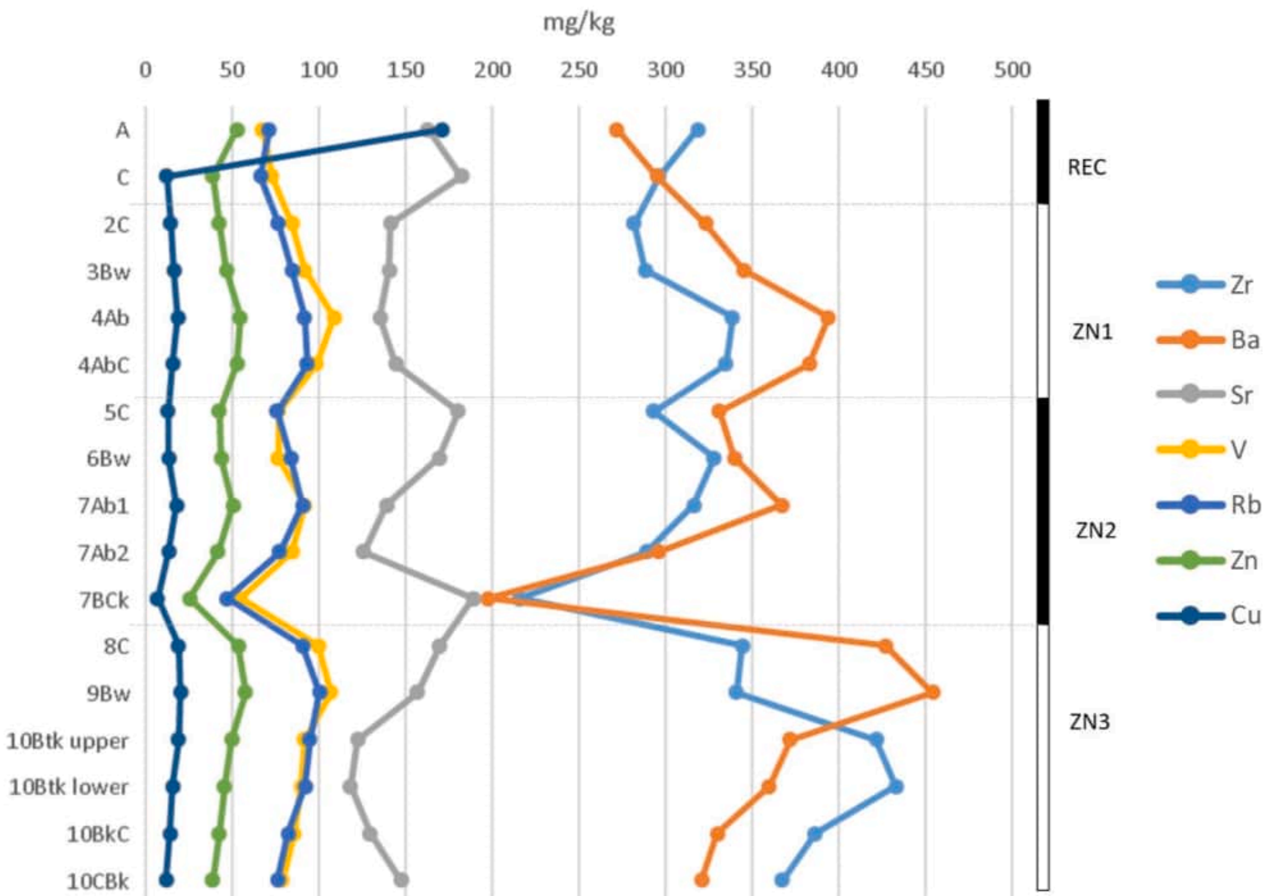

Fig. 8. Distribution of the most abundant trace elements in the sequence.

Table 5
Results of the dating of gastropod shells. The calibrated radiocarbon age before the present (BP; 0 BP = CE 1950) is quoted with a 2σ error margin.

Sample	Sampling depth (cm)	Fraction of modern		Radiocarbon age (years)		Calibrated range (yr BP)	Calibrated age (yr BP)
		pMC	1σ error	BP	1σ error		
ZN-REC C	70	18.00	0.11	13775	49	16588–16820	16704 ± 116
ZN 1 2C	392	5.630	0.070	23111	100	27253–27479	27366 ± 113
ZN 1 3Bw	446	5.186	0.062	23771	96	27760–27938	27849 ± 89

evidence of carbonate mobilization, both analytical and micromorphological. The carbonate content varies between 8 and 20 %. While it does not relate to particle size (R^2 varies between 0.0007 for carbonate-clay and 0.0234 for carbonate-fine silt), carbonate content positively correlates with $CaO + Na_2O + MgO + K_2O)/Al_2O_3$ and negatively with Ba/Sr weathering indices (Fig. 9), supporting their interpretation as partial leaching of bases, which is mostly expressed in ZN3.

On the other hand, lithologic discontinuities in the pedogenic sense at the top of each sequence are also detected by the clay-free particle size distribution and the GSI (Fig. 4), where the contents of clay and fine silt on one side, and coarse silt on the other, are mirroring each other: "younger" horizons at the beginning of the sequences (more evident in ZN2 and ZN3) have coarser fractions than the horizons at the bottom. Finally, the evident yellowish colours (2.5Y hues) of these horizons also evidence a low weathering degree (Table 1).

Clay content is highly variable, from 6 % to 34 %, also following the same trend as fine silt (increasing from the top to the bottom of each sequence). In Mediterranean loess-paleosol sequences this increase is often linked to clay illuviation during warm, wet periods, associated with rubefaction (Durn et al., 1999). In our case, the reddest horizons (10Btk, with the highest chromas, Table 1) have also high clay contents and are also the most leached (Fig. 8), but do not show any sign of clay illuviation as microlaminated clay coatings in thin sections (Table 4). The mineralogy, both bulk and clay fraction only (Table 3) shows a predominance of weatherable minerals, with a surprisingly high muscovite content in the redder horizons of ZN3, and with a fairly uniform clay composition throughout the whole profile, which rules out distinct clay neoformation in the most clayey horizons. All these observations point to true lithological discontinuities as the explanation of particle-size changes, in particular of clay, due to either different transporting agent (true aeolian / alluvial from redistributed loess), to water deposits in still, shallow environments, or to different wind speeds that generated the aeolian sequence (Újvári et al, 2016; Sümegi et al., 2019). Indeed, the clear particle sorting at the top of ZN2 (Fig. 5a) suggests an alluvial transport, either fluvial or sheet-wash type, which in turn point to the existence of shallow water tables. This is also supported by the frequent redoximorphic features in the whole sequence (Fig. 5).

The micromorphological analyses show, on one hand, that clay illuviation is not an apparent soil-forming process in this profile. On the contrary, calcitic features show a great variability. Most of the carbonates are found as calcitic micromass or as coarse elements; and the calcitic features that could be formed by leaching are represented by

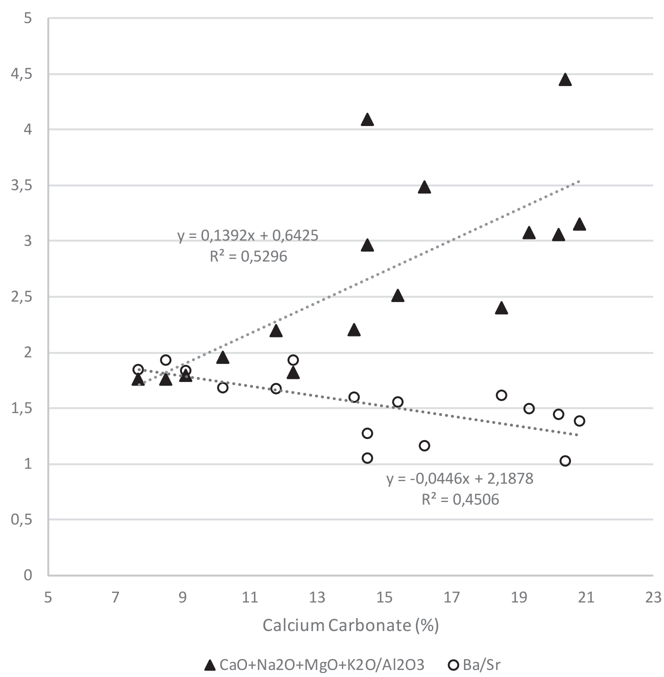

Fig. 9. Relation between $CaCO_3$ content and two geochemical indices.

impregnative micritic hypocoatings around pores and micrite nodules (Table 4, Fig. 6), more abundant in ZN3. The frequent biogenic forms of secondary calcite (earthworm spherulites, calcified roots) cannot be considered as originated by carbonate leaching since the mechanisms of formation differ from those (Álvarez et al., 2024).

In summary, the pedogenesis in the studied sequence is rather limited and periodically rejuvenated by a new supply of materials, by wind or water. The most prominent processes are a partial leaching of basic cations and a weak carbonate mobilization in the first 3 profiles, and more prominent in ZN3.

5.2. Correlation of the sedimentary units and paleoenvironmental implications

Paleoclimate and environmental reconstruction were established based on a stratigraphic framework for abrupt climatic changes during the Last Glacial period based on three synchronised Greenland ice-core records (Rasmussen et al., 2014) and multi-proxy palaeoecological studies by Molnár et al. (2010, 2011, 2021).

Sedimentation of the parent material of the youngest soil (ZN REC) corresponds with Heinrich Event 1 (16.6 ka BP), based on ^{14}C analyses of shells (Hemming, 2004). Heinrich Event 1 is followed by the Bølling–Allerød (BA) interstadial period, roughly situated between 14.7 and 12.9 ka BP. It is well-recognised in ice core data as a moderate climate period. Rasmussen et al. (2014) divide BA into sub-events (GI-1a to GI-1e). Still, the age of pedogenesis cannot be specified, since ^{14}C dates of gastropod shells indicate the time of loess sedimentation, which was settled by gastropods. Their chronology indicates sedimentation of the parent material of ZN_REC C during GS-2.1a, followed by BA pedogenesis. IRSL age of underlying pedogenetically unaltered loess corresponds to GS-2.1, indicating continuous sedimentation during Marine Isotope Stage 2 (MIS 2) (Upper Last Glacial Maximum). The ZN REC soil could be a palaeoclimatological archive of the warm BA, in a similar way as it is preserved in the Đurđevac dune sands (Galović et al., 2023b; Beerten et al., in press). If the ZN REC soil had remained unburied on the surface during the whole Holocene, the degree of pedological development would have been much higher (Galović et al., 2011, 2024) due to the more favourable conditions for weathering. Thus, following the conclusions of Galović et al. (2023a) and Beerten et al. (in press), Holocene synpedological aeolian sedimentation could occur, and slow down soil formation (MIS 1). Paleosols indeed often form through the interaction of slow deposition and climatic influences (Sümegi et al., 2018) as indicated by studies of modern pedogenesis during active loess deposition (e.i. Eger et al., 2012; Sümegi et al., 2019). On the other hand, ZN REC is situated on the loess plateau close to the cliff with strong wind erosion (Galović et al., 2023a). Thus, the possibility cannot be excluded that all the material that could have been sedimented after Heinrich Event 1 has been eroded, and that the loess deposited during Heinrich Event 1 is now exposed to pedogenesis and climatologically represents recent pedogenesis. The degree of pedogenetic development on the investigated micro-location is lower than usual in this area (Husnjak, 2014) due to synpedogenetic erosion and the supply of fresh material (Galović et al., 2023a) and due to the micro-location of the investigated profile (Galović and Peh, 2014).

The parent material of the ZN 1 section was deposited during MIS 3, after MIS 4 glaciation and sedimentation of a couple of meters thick loess in intervals from 49,900 ± 5 to 20,200 ± 2.1 ka BP (Fig. 2; Table 5). Rasmussen et al. (2014) characterise this period as an intensive alternance of cold and warm subperiods. Thus, the ZN 1 section includes three sedimentation/pedogenesis cycles. Sedimentation of loess pedogenetically altered to ZN1 4Ab and ZN1 4AbC happened during short cold periods nominated as GS-12 at the latest by GS-4 and the pedogenesis of these sediments (mainly bioturbation and organic matter accumulation) took place in the warm periods between GI-12 and at the latest by GI-4. The upper border to the 3Bw horizon is erosional and the extent of the hiatus cannot be assumed. The sedimentation of the parent material of the 3Bw and 2C horizons, which interrupted the pedogenesis of the underlying soil, occurred during the cold GS-3. The GS-3 is characterised by two dust peaks with a short warming in-between (very weak pedogenesis of the 3Bw) and a final warming corresponding to GS-2.1c (very weak pedogenesis of the 3C). This section is a palaeoclimatological and palaeoecological archive of a gradually warming period failing to cause a significant carbonate mobilisation, with a meaningful (45–50 %) shade-loving species ratio, as shown by the faunal composition of several nearby loess-paleosol sequences (Hupuczi et al., 2010; Molnár et al., 2011; Sümegi et al., 2018, 2019). In general, the milder weathering conditions and pedogenetic intensity compared to those in higher latitudes (Trájer, 2023) suggest a transition to more Mediterranean (drier) conditions in this period.

Pigati et al. (2013) investigated the reliability of radiocarbon dating late Quaternary loess deposits using small terrestrial gastropod shells. They concluded that even though they may reflect trace amounts of contamination, small terrestrial gastropod shells provide reliable ages for late Quaternary loess deposits. The difference in age of gastropods living during pedogenesis of ZN1 3Bw and in the overlying ZN 2C is 300 to 650 years. This tells us that their pedogenesis lasted for a few hundred years,

The geochronological frame of the older analysed paleosols is established based on IRSL dates (Galović et al., 2009). The ZN2 section is sedimented during the MIS 4 cold period (long-lasting GS-19.1) and its lower part is better developed than ZN 1 during intensive warming GI-18, demonstrated by organic matter accumulation and carbonate mobilisation (lower part: ZN 2 7CBk, 7BCk, 7Ab2 and 7Ab1). Pedogenesis of the 6Bw and 5C horizon most probably corresponds to GI-19.1 warming and not-so-intensive warming at the beginning of GS-18, respectively. This sequence could be described as a cool climate and noticeable wood cover (Molnár et al., 2021).

The warmer and mainly wooded ZN 3 sequence could be considered younger than 217,000 ± 22,000 years, thus it was formed probably in the MIS 7a and/or MIS 5 (Molnár et al., 2021). However, clear erosional borders in the base of the paleosol and among horizons (Galović et al., 2024), together with the degree of paleopedological development, indicate floods and mudflow redeposition of previously formed soils and sediments, that could correspond to MIS 5e warming (Eemian, around 124,000–119,000 years ago, the last interglacial period before the

Holocene, when the global mean surface temperatures were at least 2 °C warmer than now). This sequence contains the most prominent carbonate nodules that would have been developed during that period.

6. Conclusions

The pedological study confirms the presence of three distinct paleosols besides the recent soil within the Zmajevac profile. The clay-free particle size distribution also reveals significant lithological discontinuities at the beginning of each sequence. This trend, along with the presence of yellowish hues indicating lower weathering degrees, suggests that these differences are due to variations in sedimentation processes, likely influenced by changes in wind speed and transport/sedimentation agents.

Regarding pedogenesis, there is no evidence of clay illuviation typically associated with warmer and wetter periods. The mineralogy of the profile shows a predominance of weatherable minerals with consistent clay composition, indicating that the changes in clay content are more likely due to lithological discontinuities rather than pedogenic processes. Likewise, our findings show evidence of carbonate mobilization and leaching of bases. The upper sequences are characterized by a consistent decrease in the base loss index and an increase in the Ba/Sr ratio with depth, indicating a progressive degree of leaching and weathering. Carbonate accumulation is more pronounced in the deepest ZN3 profile, which probably underwent recarbonatation from calcitic loess deposited above. This is shown by carbonate content correlating with base leaching indices but not with particle size separates. Micromorphological analyses reveal most carbonates are present as micromass or coarse elements, with limited signs of leaching. Besides the frequent secondary carbonate features observed in the deepest horizon, the morphology of some of them suggest that they are primarily due to biogenic processes rather than significant pedogenic leaching.

The most recent (ZN REC) soil might serve as a paleoclimatological archive of the Bølling–Allerød interstadial period. The study suggests that the investigated profile has experienced synpedogenetic erosion and fresh material supply, resulting in lower degrees of pedogenetic development compared to other areas. Radiocarbon dating of gastropod shells further indicates that the pedogenesis of ZN1 paleosol reflects short-term climate oscillations during the Late Pleniglacial, similar to the transitions observed in ZN2 and ZN3.

Our conclusions emphasize the importance of interdisciplinary research that integrates diverse methodologies and approaches from both geology and soil science to fully understand past climatic events.

CRediT authorship contribution statement

Rosa M Poch: Conceptualization, Investigation, Methodology. **Lidija Galović:** Conceptualization, Funding acquisition, Project administration, Investigation. **Stjepan Husnjak:** Investigation, Methodology. **Jasmina Martinčević Lazar:** Writing – review & editing, Writing – original draft, Methodology, Investigation. **Nina Hećej:** Investigation, Methodology. **Stanko Ružičić:** Investigation, Methodology. **Ajka Pjanić:** Investigation, Methodology, Project administration. **Daniela Álvarez:** Investigation, Methodology. **Koen Beerten:** Investigation. **Rodoljub Gajić:** Investigation. **Petar Stejić:** Investigation. **Mihajlo Pandurov:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.catena.2024.108507.

Data availability

Data will be made available on request.

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